Scenario Projections of Infant Pertussis Burden With Vaccine Transmission Blocking and Macrolide Resistance

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Key Points

Question: How do vaccine transmission blocking and macrolide-resistant *Bordetella pertussis* dynamics alter scenario projections of infant pertussis burden?

Findings: In this deterministic transmission-modeling scenario analysis of 10 illustrative country profiles, simulated infant burden varied substantially across vaccine mechanism and resistance-management scenarios. A hypothetical high-transmission-blocking vaccine profile produced the largest reductions, whereas maternal-household composite and resistance-guided treatment scenarios were sensitive to implementation and resistance-fitness assumptions.

Meaning: Pertussis transmission models should report clinical protection, transmission blocking, infant burden, total infections, and resistant infections separately; these scenario projections should not be interpreted as national forecasts or policy rankings.

Abstract

Importance: Pertussis continues to threaten young infants despite established vaccination programs, and macrolide-resistant *Bordetella pertussis* is spreading internationally. Scenario models that combine clinical protection with transmission blocking may obscure mechanisms relevant to infant burden and antimicrobial resistance.

Objective: To evaluate how vaccine transmission-blocking assumptions and macrolide-resistant strain dynamics alter scenario projections of infant pertussis burden.

Design: Deterministic age-structured transmission-modeling scenario analysis using a 15-year burn-in and a 26-year horizon beginning January 1, 2025.

Setting and Data Sources: Ten purposively selected country profiles used population, immunization, contact, surveillance, and resistance evidence accessed through May 9, 2026.

Exposures: Vaccine mechanism profiles, resistance prevalence and fitness assumptions, current-program modifications, implementation-dependent maternal-household and resistance-guided management scenarios, and hypothetical transmission-blocking product profiles.

Main Outcomes and Measures: Annualized infant symptomatic cases, all incident infections, reported cases, resistant-strain incident infections, resistant fraction, and relative reductions.

Results: Across 10 calibrated illustrative profiles, conditional-overlay median annualized infant case incidence under current practice ranged from 15 in Thailand to 2,517 in Australia per 100,000 infants; infant incidence was not directly calibrated, so absolute rates are comparative endpoints. Overall calibration-window mean absolute percentage error for reported cases was 5.4% (IQR, 3.9%-7.8%). Stronger transmission-blocking vaccine profiles reduced infant cases more than the symptom-protective profile. Compared with current practice, the maternal-household composite transmission-reduction proxy and resistance-guided treatment produced median infant-case reductions of 44.4% and 40.5%, respectively, in the main scenario summaries; scenarios including a hypothetical high-transmission-blocking vaccine profile had reductions exceeding 95%. Resistance-management findings were sensitive to resistant-strain fitness, testing reach, and PEP assumptions.

Conclusions and Relevance: In this scenario-based transmission-modeling analysis, infant pertussis burden was sensitive to assumptions about vaccine transmission blocking and macrolide-resistant strain dynamics. These findings support reporting clinical protection and onward transmission effects separately in pertussis models and should be interpreted as scenario projections rather than national forecasts or policy recommendations.

Introduction

Pertussis continues to cause substantial morbidity among young infants despite established childhood vaccination programs and maternal immunization strategies in many settings.^1-3^ Current policy debates include maternal immunization, booster strategies, improved vaccines, treatment, postexposure prophylaxis (PEP), and antimicrobial resistance.^4-6^ Interpreting the potential effect of new or modified vaccination strategies is challenging because vaccines may reduce symptoms, susceptibility, infectiousness, and duration of infectiousness to different degrees. This distinction is particularly important when infant protection depends not only on direct immunity but also on reduced transmission from older age groups.

Post-pandemic pertussis resurgence has occurred despite high routine childhood vaccination coverage, consistent with models in which waning and immune boosting shape long-term dynamics.^7,8^ Acellular pertussis (aP) vaccines can prevent disease while providing incomplete protection against infection and onward transmission.^9^ Epidemiologic evidence and meta-analyses similarly indicate that aP-derived protection against infection wanes substantially within 3 to 5 years, while protection against severe disease persists longer.^10-13^

Simultaneously, macrolide-resistant *B pertussis* (MRBP) has emerged as an international concern. Shanghai genomic surveillance found macrolide resistance rose from 50% or less before 2020 to nearly 100% after 2020, linked to the 23S rRNA A2047G mutation; in a 2024 multicenter China outbreak study, 99.5% of isolates belonged to an MT28-Shanghai clone carrying ptxP3, A2047G, and prn150.^2,14,15^ MRBP and related resistant lineages have also been reported in public health summaries for the Americas and other settings.^16,17^ Genomic reports from Australia and Japan, together with global spread analyses, provide additional evidence of international dissemination.^18-21^ This rapid dissemination challenges the assumption that resistant strains necessarily carry a major transmission disadvantage and raises the prospect that macrolide treatment and postexposure prophylaxis (PEP) may become progressively less effective.

Most pertussis policy models report vaccine effectiveness as a single composite endpoint and do not explicitly model how transmission-blocking properties interact with antimicrobial resistance. We therefore used 10 purposively selected country profiles as contrasting policy-relevant scenarios, not as a representative global sample, to evaluate how alternative vaccine-mechanism and resistance assumptions alter simulated infant burden, total infections, reported cases, and resistant infections under specified assumptions.

Methods

Design and Data Sources

This deterministic transmission-modeling scenario analysis synthesized demographic, surveillance, immunization, contact, treatment, and resistance evidence in an age-structured pertussis framework. Current practice was defined as each profile's routine vaccination schedule and standard macrolide treatment/PEP assumptions. The analysis was designed to compare epidemiologic scenario projections under specified assumptions, not to forecast national epidemics or estimate cost-effectiveness, feasibility, equity, or a complete policy appraisal. Reporting followed relevant non-cost CHEERS 2022 elements and WHO immunization-modeling guidance.^22,23^ Institutional review board review was not applicable because no individual-level data were used.

Ten purposively selected country profiles were analyzed: Australia, Brazil, China, Japan, New Zealand, South Africa, Sweden, Thailand, the United Kingdom, and the United States. The profiles are illustrative settings chosen to span program, surveillance, contact, and resistance contrasts, not to estimate region-level or global averages. Inputs included United Nations World Population Prospects 2024 denominators, WHO/UNICEF and national immunization records, Prem/contactdata contact matrices, harmonized surveillance intervals, and resistance evidence accessed through May 9, 2026.^24-27^ Country-selection rationale and data-quality dimensions are reported in eTable 1.

Model Structure and Calibration

The model tracked 8 age groups, 2 strains, and susceptible origin histories reflecting maternal protection and recent or waned dose histories. Four vaccine mechanism parameters were represented separately: VE_sus, reduced susceptibility to infection; VE_sym, reduced symptoms given infection; VE_inf, reduced onward infectiousness; and VE_dur, shortened infectious duration. Immunity followed a SIRWS structure with immune boosting, waning from recovered to partially immune status over approximately 5 years, and return to susceptibility over approximately 10 years.^7,8^ Transmission used country-specific contact matrices, seasonality, aging and birth turnover, vaccination maintenance, importation, COVID-19 non-pharmaceutical intervention (NPI) contact reductions, treatment, and PEP. Table 1 classifies the main model components by analytic role, source, and uncertainty handling; eTable 2 provides the full study parameter-design matrix with source/provenance links, and the Supplement provides equations, diagrams, transition tables, vaccination schedules, and parameter details.

Calibration targeted reported pertussis case counts over the 6 most recent observed calendar years in each harmonized surveillance series, except South Africa, where 3 overlapping years were available. The retained fit minimized a negative-binomial reporting-interval likelihood plus penalties for age-specific reporting probabilities outside literature-informed bounds. Model-data checks included observed-vs-modeled reporting intervals, reporting probabilities, fit scores, mean absolute percentage errors, peak timing errors, and peak magnitude ratios (Figure 1C, eFigure 2B and 2D, and eTables 7 and 12). Consistent age-specific observed incidence series were unavailable across all 10 profiles, so age-specific incidence was not a calibration target; resistance hindcasts were treated as plausibility checks rather than held-out validation.

Calibrated parameters included country-specific transmission β_S and reporting multipliers, with age-specific reporting probabilities constrained by broad prior bands; natural-history, contact, vaccine-mechanism, and resistance parameters were fixed or varied in scenarios and sensitivity analyses. Because infant incidence was not directly calibrated, infant outcomes are internally consistent scenario endpoints rather than externally validated infant forecasts. Credibility checks relied on reported-incidence fit, fitted infant reporting gradients, separate 0-2 month and 3-11 month strata, infant-contact sensitivity, age-shift diagnostics, and infant-age-stratified intervention-window diagnostics (eTables 7, 12, 17, 18, and 22). Consistent country-specific hospitalization or severity series were not available for formal calibration.

Scenarios and Outcomes

Five vaccine profiles were compared: no vaccine, symptom-protective aP-like protection, infection-blocking protection, transmission-blocking protection, and a hypothetical high-transmission-blocking profile motivated by upper-bound mucosal-immunity product targets rather than an available vaccine (eTable 2).^28,29^ Resistance scenarios used country-specific timelines or fixed low-to-very-high resistant prevalence, then allowed resistant dynamics to depend on importation, relative fitness, treatment shortening, strain-specific PEP effectiveness, and age/contact transmission.

Interventions included marginal child-coverage increases in already high-coverage profiles, adolescent boosting, a maternal-household composite transmission-reduction proxy, resistance-guided treatment under a main implementation scenario, the hypothetical high-transmission-blocking vaccine profile, and a combined strategy.^30^ We grouped scenarios as current-program modifications, implementation-dependent management or proxy packages, and hypothetical product-target vaccine profiles. The intervention set intentionally mixed available or near-available levers, implementation-dependent proxy packages, and a hypothetical product-target vaccine scenario; therefore, scenarios were grouped by interpretability rather than treated as mutually substitutable policy options. The maternal-household composite transmission-reduction proxy combined passive newborn antibody protection, a reproductive-age adult boosting proxy, and a cocooning contact modifier; decomposition runs isolated these components.

The primary analysis compared current practice with predefined interventions for annualized infant symptomatic cases; vaccine transmission-blocking profiles and resistance-fitness interactions were key mechanistic analyses. Order-stability summaries, selected-parameter joint parameter-sampling diagnostics, QALY-like translation, the stochastic toy model, and vaccine-pipeline mapping were exploratory diagnostics.

Infant outcomes combined the 0-2 month and 3-11 month age groups, with the 2 strata retained separately for reporting-probability, age-shift, and intervention-window diagnostics. Symptomatic cases were incident infections after VE_sym modification; reported cases applied age- and country-specific reporting probabilities; all infections included incident symptomatic and asymptomatic infections; resistant infections were incident infections caused by the resistant strain. Annualized outcomes were averaged over the 26-year saved analysis horizon unless otherwise specified. Relative reduction was defined as $1 - Z/Z_0$.

Sensitivity Analysis

Sensitivity analyses included vaccine-mechanism thresholds, resistant-fitness grids, reporting assumptions, resistance-mechanism decomposition, treatment and PEP implementation scenarios, infant contact-matrix multipliers, maternal passive-protection duration, burn-in and COVID-19 NPI temporal assumptions, marginal child-coverage mechanism diagnostics, scenario order across windows and infant strata, event-scale diagnostics, selected-parameter joint order-stability diagnostics, a contactdata setting-specific stochastic toy model, QALY-like burden translation, and vaccine-pipeline mapping. A 48-run Latin-hypercube analysis used correlation diagnostics for exploratory parameter screening, not variance decomposition; a separate 128-sample selected-parameter joint order-stability diagnostic varied reporting, infant contacts, VE_inf, resistant fitness, asymptomatic transmission, resistance-management uptake, and PEP reach. Conditional beta-grid intervals propagated transmission-rate uncertainty over an adaptive log-β_S grid with nuisance parameters fixed; throughout, these are conditional intervals and do not jointly propagate structural, resistance-fitness, contact-matrix, or implementation uncertainty.

Results

Simulated Burden Under Current Practice

The 10 illustrative calibrated country-profile scenarios produced wide variation in simulated burden under current aP-like vaccination and country-specific resistance anchors (Figure 1C and 1D). In the saved-horizon comparison plotted in Figure 1C, model-reported incidence differed from recent observed mean reported incidence by a median of 65.0% (IQR, 14.1%-143.4%; range, 6.2%-350.3%); this horizon-scale comparison is not identical to the interval-level calibration target. Direct calibration-window diagnostics showed mean absolute percentage error of 5.4% (IQR, 3.9%-7.8%; range, 1.2%-10.5%), median absolute peak-timing error of 1 year, and variable peak-magnitude ratios, underscoring that the fits support scenario comparison rather than outbreak forecasting (Table 2 and eTable 7). Within these calibrated profiles, conditional-overlay median annualized infant case incidence ranged from 15 per 100,000 infants in Thailand to 2,517 in Australia, all-infection incidence from 33 to 3,192 per 100,000 persons, and reported incidence from 0.5 to 53.4 per 100,000 persons (Figure 1D). Because infant-specific incidence was not a calibration target, absolute infant rates should be read primarily as internally consistent comparative endpoints. Near-term checks showed sensitivity to burn-in duration and, in Australia, NPI contact-shock magnitude (eTable 21).

Infant-Outcome Robustness Diagnostics

Because infant incidence was not calibrated directly, the infant endpoint is the main calibration vulnerability; Table 3 therefore reports 0-2 month and 3-11 month scenario contrasts across analysis windows. In the contact-matrix sensitivity, increasing child, adolescent, and adult sources into infant target groups from 0.75 to 1.50 changed current-practice 5-year median infant burden from 145.8 to 228.3 cases per 100,000 infants; the maternal-household composite transmission-reduction proxy remained lower across the same multipliers (93.4 to 138.5 per 100,000 infants). Varying passive maternal protection from 90 to 270 days changed direct-antibody-only 5-year reductions from 2.7% to 4.3% and the maternal-household composite transmission-reduction proxy from 55.1% to 56.8% (both in eTable 17).

Across 5-, 10-, and 26-year windows, median reductions were 57.6%, 49.5%, and 43.1% for the maternal-household composite transmission-reduction proxy and 15.9%, 43.7%, and 40.7% for resistance-guided treatment, respectively (eTable 19). In the 128-sample joint order-stability diagnostic varying reporting multiplier, infant contacts, VE_inf, resistant fitness, asymptomatic transmission, resistance-management uptake, and PEP reach, the lowest simulated infant burden occurred in combined (69.5%) or hypothetical high-transmission-blocking vaccine (30.5%) scenarios, while maternal-household composite and resistance-guided treatment scenarios remained intermediate (eTable 25). These diagnostics support broad separation among mechanism categories but do not externally validate infant forecasts.

Vaccine Transmission-Blocking Mechanism Scenarios

Vaccine profiles with stronger infection or transmission effects yielded progressively lower simulated infant and population burden (Figure 2A, 2B, and 2D). Compared with no vaccination, the symptom-protective aP-like profile produced an 82.0% simulated infant-case reduction, whereas infection-blocking and hypothetical high-transmission-blocking profiles reduced residual infant burden to very low deterministic levels in several profiles (Figure 2B); these low levels should not be interpreted as stochastic elimination probabilities. All-infection outcomes showed similar directional patterns (Figure 2D). Vaccine-history decomposition showed that infections under the current aP-like profile remained concentrated in dose-3-or-more and waned histories, whereas residual infections under the hypothetical high-transmission-blocking profile were more concentrated in unvaccinated histories (Figure 2C). Threshold analyses suggested that VE_inf values near 0.35 to 0.50 were needed to cross selected infant-case reduction or comparator thresholds when other vaccine mechanisms were held fixed (eTable 14).

Macrolide-Resistance Sensitivity Analyses

Resistant infection burden was sensitive to initial resistance prevalence, treatment pressure, PEP effectiveness, resistant-strain fitness, and vaccine infectiousness effects (Figure 3A-3F). Under neutral-fitness country-timeline assumptions, the model generated a high median end resistant fraction (99.7%, from 1.0%; Figure 3A), but this result should be interpreted as a stress-test output under specified treatment and PEP assumptions rather than a forecast of inevitable resistant-strain replacement. Mechanism decomposition suggested that resistant fitness and treatment/PEP differentials drove most divergence: equalizing PEP effectiveness lowered the median end resistant fraction to 12.2%, equalizing both treatment and PEP effects lowered it to 1.0%, and imposing $f_R = 0.85$ lowered it to 0.01% (Figure 3C and eTable 13). In the resistant-fitness sensitivity and fitness-by-VE_inf grid, the median end resistant fraction was less than 0.1% with $f_R = 0.85$, 99.7% with $f_R = 1.00$, and 99.9% with $f_R = 1.15$ (Figure 3B and 3D). Increasing VE_inf from 0.05 to 0.55 produced lower simulated infant burden across fitness assumptions, but the magnitude was fitness-dependent (Figure 3E and 3F).

Scenario Projections by Intervention Category

Scenario projections were grouped by interpretability rather than presenting them as mutually substitutable policy options (Figure 4A-4C). Current practice had median annualized infant case incidence of 341 per 100,000 infants/year (Figure 4A and 4D). Among current-program modification scenarios, marginal child-coverage increases in already high-coverage profiles yielded no consistent median infant-case reduction (-4.2%; IQR, -4.7% to -3.4%; Figure 4B). Mechanism diagnostics suggested small age-shift and waning-immunity composition changes under specified assumptions; this unstable model response should not be interpreted as evidence against maintaining high routine childhood coverage, which remained a prerequisite baseline assumption in these scenarios (eTable 18). This result reflects marginal changes in already high-coverage profiles, not the value of routine childhood vaccination itself. Adolescent boosting produced a small median simulated reduction (1.9%; IQR, -0.2% to 58.4%; Figure 4B).

Among implementation-dependent management or proxy scenarios, the maternal-household composite transmission-reduction proxy produced a simulated 44.4% infant-case reduction under specified implementation assumptions. Decomposition attributed a median 7.4% reduction to direct antibody protection alone, 34.7% to the reproductive-age adult boosting proxy, and 6.4% to cocooning; therefore, the proxy estimate should not be interpreted as passive maternal antibody protection alone. Infant contact-matrix sensitivity confirmed that adult-to-infant contact assumptions influenced absolute infant burden (eTable 17). Resistance-guided treatment produced simulated reductions of 40.5% for infant cases and 96.7% for resistant infections in the main implementation scenario, but near-term implementation checks showed dependence on uptake, testing reach, PEP restoration, and PEP reach (eTable 16).

Among hypothetical product-target vaccine profiles, the high-transmission-blocking vaccine and combined stress-test scenarios produced simulated infant-case reductions greater than 95%; the vaccine scenario represents a product target rather than an available intervention. Near-term analysis-window checks retained very large point-estimate reductions for scenarios including the hypothetical high-transmission-blocking vaccine profile but changed the relative positions of the maternal-household composite transmission-reduction proxy, adolescent-booster, and resistance-guided-treatment scenarios (eTable 19). Exploratory infant-stratum, event-scale, order-stability, and stochastic toy-model diagnostics are summarized in the Supplement and support interpreting very low deterministic burdens as low-burden thresholds rather than stochastic elimination probabilities (eTables 20, 22, 23, 25, and 26).

Discussion

In this deterministic scenario-modeling analysis across 10 illustrative country profiles, projected infant pertussis burden varied substantially according to assumptions about vaccine transmission blocking and macrolide-resistant strain dynamics. Scenarios that reduced onward transmission generally produced larger simulated infant-burden reductions than scenarios that primarily increased clinical protection. Resistance-management scenarios also altered both infant burden and resistant infections, but these results depended strongly on assumptions about resistant-strain fitness, diagnostic capacity, and PEP effectiveness.

The distinction between clinical protection and transmission blocking had quantifiable consequences. A symptom-protective aP-like profile produced lower simulated infant cases than no vaccination, but stronger transmission-blocking profiles produced larger simulated decreases in infant cases and total infections. This is consistent with evidence that aP vaccination prevents disease more strongly than colonization or transmission,^9^ and suggests that models using a single composite vaccine-effectiveness parameter may underestimate the hypothetical simulated benefit of vaccines designed to reduce onward transmission.^28,29^

Resistance-guided treatment had simulated benefit under the main implementation assumptions, aligning with guidance emphasizing susceptibility testing and alternative treatment when resistance is suspected or confirmed.^16,17,30^ However, near-term implementation scenarios showed that simulated benefit depended on uptake, testing reach, and PEP assumptions; real-world benefit would depend on diagnostic capacity, rapid turnaround, clinician suspicion, alternative-drug availability, adherence, tolerability, effective PEP implementation, and public health capacity.

The simulated increase in resistant fraction should be interpreted as conditional rather than deterministic. In particular, high resistant fractions under neutral-fitness assumptions represent stress-test outputs under specified treatment and PEP assumptions, not predictions of inevitable resistant-strain replacement. Hindcast plausibility checks did not rule out neutral or above-neutral fitness in China, Japan, and Australia, but they should not be read as validating a unique fitness estimate. A conservative $f_R = 0.85$ counterfactual produced a median end resistant fraction below 0.1% in the fitness sensitivity and 0.01% in the mechanism-decomposition run, highlighting that resistance conclusions depend on fitness and management assumptions.

Scenario contrasts depended on the interaction between vaccine mechanism and resistance. Marginal child-coverage increases in already high-coverage profiles did not produce a consistent median simulated infant-case reduction, and age-shift diagnostics suggested that the small modeled increase was not a robust policy signal or evidence against maintaining high coverage. The maternal-household composite transmission-reduction proxy requires careful interpretation because the adult-boosting component should not be attributed to passive transplacental antibody protection alone, and infant-contact sensitivity analyses indicate that household-like contact assumptions remain important for infant outcomes. This scenario should therefore be interpreted as the maternal-household composite transmission-reduction proxy rather than an empirical estimate of pregnancy vaccination effectiveness.

Limitations

This study has limitations that should guide interpretation of the scenario projections. First, infant incidence was not directly calibrated because consistent age-specific observed incidence targets were unavailable across the 10 profiles. Absolute infant-burden estimates are therefore less secure than internally consistent contrasts in scenario direction, although fitted reporting gradients, separate infant strata, infant-contact sensitivity, and age-window diagnostics probed this vulnerability.

Second, intervals are conditional and do not include full structural or implementation uncertainty. The beta-grid intervals propagate transmission-rate uncertainty with nuisance parameters fixed; they do not jointly propagate contact-matrix uncertainty, resistance-fitness uncertainty, implementation heterogeneity, or alternative structural assumptions. Deterministic compartmental dynamics also do not represent stochastic extinction, household clustering, contact tracing, adherence, or superspreading.

Third, resistance projections are sensitive to resistant-strain fitness, treatment, and PEP assumptions. The high resistant fractions under neutral-fitness assumptions should be read as stress-test outputs rather than forecasts of inevitable resistant-strain replacement. Fourth, the 10 countries are illustrative rather than representative; cross-country contrasts should not be read as national forecasts, global estimates, or policy rankings. Additional interpretation limits include age-structured proxies rather than explicit pregnant-person or household compartments, a hypothetical high-transmission-blocking vaccine profile rather than an available product, and exploratory QALY-like translations without costs or formal decision thresholds (eTables 24 and 27).

Conclusions

Across 10 purposively selected illustrative country profiles, deterministic scenario-based simulations of infant pertussis burden were sensitive to assumptions about vaccine effects on infection and onward transmission and to macrolide-resistant strain dynamics. These findings support reporting vaccine clinical protection, transmission effects, infant burden, total infections, and resistant infections separately in pertussis transmission models; they should be interpreted as scenario projections rather than national forecasts, cost-effectiveness estimates, or policy recommendations.

Supplement

Supplement 1. eMethods (full model equations, country-profile construction, calibration methods, scenario definitions, resistance hindcast plausibility checks, parameter identifiability classification), eReferences, eFigures 1-8, and eTables 1-28.

Article Information

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Author Contributions: Kangguo Li and Tianmu Chen had full access to all the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. Kangguo Li conducted and is responsible for the data analysis. Concept and design: Kangguo Li. Acquisition, analysis, or interpretation of data: Yulun Xie, Yunzhi Zenghuang, Tao Chen, Sha Chen, and Yue He. Drafting of the manuscript: Kangguo Li and Yulun Xie. Critical revision of the manuscript for

important intellectual content: Kangguo Li. Statistical analysis: Kangguo Li. Administrative, technical, or material support: Kangguo Li and Tianmu Chen. Supervision: Tianmu Chen.

Reporting Guideline: Relevant non-cost elements of CHEERS 2022 and WHO immunization-modeling guidance were followed; model equations, parameter classification, calibration diagnostics, and sensitivity analyses are provided in the Supplement.

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Tables

Table 1. Key Model Components and Interpretation.

Component	Role in analysis	Source or assumption	Uncertainty propagated?	Interpretation
Transmission rate β_S	Calibrated country-specific parameter	Reported-incidence series and literature-informed prior	Yes, conditional beta-grid intervals	Anchors overall reported-incidence level
Reporting multiplier and age reporting probabilities	Calibrated or penalized observation-model parameters	Broad literature-informed reporting bands	Partly; varied in reporting and selected-parameter diagnostics	Converts infections to reported cases; not a direct burden measure
Contact matrix C_{ij}	Fixed country mixing input	Prem/contactdata matrices aggregated to 8 model age groups	Partly; infant-contact multipliers and stochastic toy model	Defines age mixing and infant exposure routes
Vaccine symptom protection	Fixed or scenario-defined mechanism	Literature-informed aP-like profile and mechanism scenarios	Partly; mechanism thresholds and selected-parameter diagnostics	Represents clinical protection given infection
Vaccine infectiousness and duration effects	Scenario-defined mechanism	Hypothetical transmission-blocking profiles and product-target assumptions	No full structural propagation	Represents onward transmission blocking rather than clinical efficacy alone
Resistant-strain fitness	Sensitivity or scenario parameter	Assumed grid, $f_R = 0.70 - 1.25$, with hindcast plausibility checks	No in primary intervals	Major determinant of resistant-fraction trajectories
Treatment and PEP effectiveness	Scenario and sensitivity parameters	Standard macrolide and resistance-guided management assumptions	No in primary intervals; varied in implementation checks	Drives management benefit and resistant-strain selection pressure

Component	Role in analysis	Source or assumption	Uncertainty propagated?	Interpretation
Infant incidence	Model-generated endpoint	Not directly calibrated to age-specific observed incidence	No external calibration	Primary comparative endpoint, not a validated national forecast

Table 2. Country-Specific Calibration Diagnostics.

Country	Intervals	Observed cases	Modeled cases	MAPE, %	Peak timing error, y	Peak magnitude ratio
Australia	65	88,963	88,963	1.8	1	0.93
Brazil	64	11,275	11,275	8.5	-3	0.17
China	81	640,783	675,181	3.9	1	0.57
Japan	276	82,325	82,325	4.8	0	0.45
New Zealand	64	5,324	5,324	3.9	2	0.69
South Africa	32	3,883	3,883	1.2	-2	0.60
Sweden	64	3,562	3,562	10.5	1	0.24
Thailand	72	1,982	1,982	7.0	0	0.22
United Kingdom	273	34,581	34,581	8.1	0	0.21
United States	278	25,679	25,679	6.1	-3	0.22

Table 3. Infant-Stratum Scenario Contrasts Across Analysis Windows.

Window	Infant stratum	Scenario	Median cases per 100,000/y (IQR)	Median reduction vs current
5 y (2025-2029)	0-2 mo	Current practice	251.7 (55.9-778.8)	Reference
5 y (2025-2029)	0-2 mo	Maternal-household composite transmission-reduction proxy	116.9 (9.8-387.6)	69.4%
5 y (2025-2029)	0-2 mo	Resistance-guided treatment	300.3 (28.1-1277.4)	16.1%
5 y (2025-2029)	0-2 mo	High-transmission-blocking product target	1.9 (0.6-394.2)	97.8%
5 y (2025-2029)	3-11 mo	Current practice	151.0 (34.9-460.9)	Reference

Window	Infant stratum	Scenario	Median cases per 100,000/y (IQR)	Median reduction vs current
5 y (2025-2029)	3-11 mo	Maternal-household composite transmission-reduction proxy	104.5 (8.7-344.3)	52.9%
5 y (2025-2029)	3-11 mo	Resistance-guided treatment	181.5 (17.6-792.1)	15.9%
5 y (2025-2029)	3-11 mo	High-transmission-blocking product target	1.0 (0.3-225.9)	98.1%
26 y (2025-2050)	0-2 mo	Current practice	533.6 (51.0-1553.1)	Reference
26 y (2025-2050)	0-2 mo	Maternal-household composite transmission-reduction proxy	239.1 (8.0-927.0)	59.9%
26 y (2025-2050)	0-2 mo	Resistance-guided treatment	328.9 (24.2-1195.7)	41.0%
26 y (2025-2050)	0-2 mo	High-transmission-blocking product target	2.0 (0.6-558.7)	97.3%
26 y (2025-2050)	3-11 mo	Current practice	317.3 (31.9-955.3)	Reference
26 y (2025-2050)	3-11 mo	Maternal-household composite transmission-reduction proxy	212.8 (7.2-812.7)	36.3%
26 y (2025-2050)	3-11 mo	Resistance-guided treatment	198.2 (15.1-744.4)	40.6%
26 y (2025-2050)	3-11 mo	High-transmission-blocking product target	1.0 (0.4-317.7)	97.6%

For Table 3, cases are empirical medians and IQRs across the 10 illustrative profiles; reductions are country-matched median reductions vs current practice.

Figure Legends

Figure 1. International Context, Country Selection, and Baseline Heterogeneity. (A) WHO regional reported pertussis incidence for context, not global representativeness. (B) Country profile dimensions. (C) Observed vs modeled annual reported incidence; dashed line indicates equality and color indicates resistant infection fraction. (D) Baseline all infections, reported cases, and infant cases per 100,000 on log scale. Conditional beta-grid intervals are not full structural or implementation uncertainty.

Figure 2. Vaccine Mechanism Scenarios. (A) VE_sus, VE_sym, VE_inf, and VE_dur values for 5 profiles. (B) Infant-case burden by vaccine scenario and country. (C) Vaccine-history origin decomposition by profile. (D) All-infection burden by vaccine scenario and country. Panels B and D show country points, cross-country medians, and empirical intervals.

Figure 3. Macrolide Resistance and Vaccine Transmission Blocking. (A) Five-year resistant-fraction dynamics by country. (B) Fitness-dependent resistant-fraction dynamics for $f_R = 0.85 - 1.15$. (C) Infant burden across resistance scenarios. (D) End-period resistant fraction by f_R and VE_inf. (E) Infant disease burden by f_R and VE_inf on log10 scale. (F) Transmission-blocking benefit by country at 3 fitness levels.

Figure 4. Scenario Projections by Intervention Category. Scenarios are grouped by interpretability rather than ordered as mutually substitutable policy options. (A) Infant-case burden by scenario and country; points are countries, with cross-country medians and empirical intervals. Current practice is the status quo baseline comparator and is grouped with current-program modifications, including higher child coverage and adolescent boosting; other scenarios are grouped as implementation-dependent proxy or management scenarios and hypothetical product-target or stress-test scenarios. (B) Within-country percentage infant-case reduction vs current practice; cell text gives the point estimate and conditional 95% interval. (C) Median burden across infant cases, reported cases, and all infections, showing outcome-dependent scenario contrasts. (D) Baseline infant case incidence with conditional beta-grid intervals. Conditional intervals are not full structural or implementation uncertainty.